

Travel Booking AI Plan Analysis - PRD

Product Name: Travel Booking AI Assistant Framework

Document Status: Draft / MVP Phase (Updated Specifications)

Timeline: 1.5-Week Sprint (10 Days)

Crucial Scope Update & Developer Disclaimer:

The development team will **not** be responsible for live API integrations or payment gateway functionality. We are delivering a **framework product** equipped with the necessary endpoints. The client/system administrator will be required to add the respective API keys to activate external functionalities. Furthermore, the platform is restricted strictly to a **Web Application-based chatbot**. Payment gateway functionality will not be shipped with this framework.

1. Product Overview & Objective

1.1 Problem Statement

Travel planning currently requires users to bounce between 5+ applications (OTA search, hotel booking, cab aggregators, payment apps, and email for tickets). This context-switching creates friction.

1.2 Solution

An AI-first travel booking framework that allows users to search and orchestrate itineraries entirely through natural conversation on a **Web Application Chatbot**. The AI orchestrates the initial steps, eliminating complex UIs.

1.3 Business Value

- Validates an end-to-end booking flow in a conversational web interface.
- Provides a scalable architecture for future B2B/Corporate integrations.
- Delivers a foundational framework ready for API key plug-and-play.

2. Target Audience & Personas

- **Primary (MVP):** Individual Travelers looking for a quick, frictionless way to discover standard itineraries via web chat.
- **Secondary (Phase 2):** Travel Agents managing bulk bookings.
- **Tertiary (Phase 3):** Corporate employees needing automated policy-compliant bookings.

3. Core Features (MVP Scope)

- **Natural Language Search:** Multi-modal search for flights, hotels, and cabs using conversational prompts.
- **Framework Orchestration:** Structured endpoints ready to process API queries once the operator provisions their keys.
- **Web-based Ticketing Prep:** Automated generation of PDF tickets and vouchers delivered directly in the web chat (pending integration of live APIs by operator).
- **State Management:** Continuous conversation context so users don't have to repeat preferences.

4. User Journey

1. **Initiation:** User messages the web application chatbot with a travel request (destination, dates, budget).
2. **Flight Selection:** AI structures the query for flight search (via operator-provided API), presents top options. User selects one.
3. **Hotel Selection:** AI automatically structures hotel queries for the destination/dates. User selects one.
4. **Cab Selection:** AI offers airport transfers. User selects option.
5. **Checkout Handoff:** AI summarizes the total trip cost. *(Note: Payment gateway is excluded from this framework; handoff to external or manual payment completion is required).*
6. **Confirmation:** Upon mock/external confirmation, the framework replies with placeholder or externally-generated PDF tickets.

5. API Requirements & Framework Architecture

This section outlines the endpoints provided in the framework. **No live third-party API connections are shipped natively.**

5.1 Framework Endpoints provided (MVP Phase)

- **Conversational Interface:** Web Application Webhooks for inbound and outbound messaging (text, cards, PDFs). *(External messaging app integration is completely out of scope).*
- **AI & Orchestration:** Core intelligence engine endpoints ready to connect to ANY LLM once the operator supplies their API keys.
- **Mock/Placeholder Endpoints (To be mapped to Live APIs by client):**
 - **Flight API Endpoint:** Ready to receive Amadeus/Skyscanner credentials.
 - **Hotel API Endpoint:** Ready to receive Booking.com/Agoda credentials.
 - **Cab API Endpoint:** Ready to receive Uber/Local aggregator credentials.

6. Tech Stack Summary

- **Backend Runtime:** Node.js (v20 LTS) with Express.js (HTTP server / Webhooks).
- **Database:** PostgreSQL with Prisma ORM.
- **Document Generation:** PDFKit / Puppeteer (for generating tickets natively in the web app).
- **Deployment:** AWS EC2/RDS or Railway/Render.

7. Out of Scope / Future Roadmap

- **Live API Integrations:** Implementation of live Amadeus, Booking.com, or LLM APIs is the responsibility of the operator via key insertion.
- **Payment Gateways:** Integration of Razorpay, Stripe, or any financial transaction processor.
- **External messaging bot functionality.**
- Travel Agent portal for bulk CSV bookings.
- Corporate travel policy rules engine.
- Upselling logic (Travel insurance, tour packages).